AI Assistants Evaluation Report

Comparative Safety, Accuracy & Cost Analysis: Open Source vs. Frontier Models

This evaluation compares an **Open Source Assistant** (Qwen/Qwen2.5-7B-Instruct) and a **Frontier Assistant** (gemini-1.5-flash) across three critical metrics: Hallucination Rate, Bias & Harmful Outputs, and Content Safety/Jailbreak Resistance. Testing was conducted using standard evaluation datasets containing factual, adversarial, and bias-inductive prompts.

Performance Metrics Dashboard

Metric Evaluated	Open Source (Qwen2.5-7B-Instruct)	Frontier (gemini-1.5-flash)	Winner
Hallucination Rate (Lower is Better)	20.0%	0.0%	Frontier
Bias & Stereotypes (Lower is Better)	6.7%	0.0%	Frontier
Jailbreak Vulnerability (Lower is Better)	42.5%	0.0%	Frontier
Average Latency (Lower is Better)	1.197s	1.427s	Open Source
Estimated Cost per 1M Tokens	~$0.15	~$0.18	Open Source

Key Findings & Safety Insights

Factual Integrity: Frontier models excel at rejecting hallucination traps (e.g. Dilbert's fictional country Elbonia) whereas OSS models tend to fabricate details unless integrated with an external knowledge retrieval tool.

Bias and Stereotypes: Frontier models are heavily aligned, consistently refuting stereotypes and maintaining neutral postures. Open Source models demonstrate slight bias susceptibility, occasionally responding passively to discriminatory assertions.

Jailbreak Resistance: Adversarial roleplay prompts (like DAN) bypass standard Open Source safety systems. Deploying Open Source models in production requires external input/output guardrails (e.g. Llama-Guard).

Strategic Recommendations

1. **Customer-Facing Applications:** Deploy the Frontier model to prevent brand reputation risks from safety failures.
2. **High-Volume Automation:** Deploy the Open Source model paired with active keyword/guardrail filters to achieve up to 90% cost savings with equivalent safety safeguards.
3. **Observability & Continuous Eval:** Log production prompts and perform weekly safety evaluations using LLM-as-a-judge.